Tong-Yu Lin

Phone: +86 18240029769 Email: lin-ty21@mails.tsinghua.edu.cn Homepage: ltytsinghua.github.io
Ph.D. candidate in Condensed Matter Theory, Department of Physics, Tsinghua University

Education

Tsinghua University Ph.D. in Physics, Advisor: Prof. Guang-Ming Zhang **2021.9–2026.12 (expected)**
Nanjing University B.S. in Physics, Top-notch Undergraduate Program **2017.9–2021.6**
University of California, Berkeley Exchange Student, Department of Physics **2019.8–2019.12**

Research Interests

Current

- **Tensor-network methods**: developing tensor-network methods for problems in statistical physics.
- **Frustrated magnetism**: exotic magnetic orders and phase transitions in frustrated magnetic systems.
- **Phenomenology of unconventional superconductors**: unconventional vortex excitations in superconductors, such as fractional vortices.

Future

- **Physics for AI**: improving the reasoning and physics capabilities of AI, and studying AI interpretability.
- **AI for Science**: using AI to accelerate discovery in physics.

Publications

Feng-Feng Song, **Tong-Yu Lin** and Guang-Ming Zhang, General tensor network theory for frustrated classical spin models in two dimensions, Phys. Rev. B 108, 224404 (2023) **Co-first author**

• The ground states of frustrated classical spin models are highly degenerate, so standard tensor-network representations fail to converge at low temperatures. We proposed a general tensor-network theory that encodes the emergent degrees of freedom into local tensors to relieve frustration; the resulting tensor networks converge and yield accurate physical results.

Tong-Yu Lin, Feng-Feng Song and Guang-Ming Zhang, Theory of the charge-6e condensed phase in kagome-lattice superconductors, Phys. Rev. B 111, 054508 (2025) **First author**

• Proposed a phenomenological theory for the nontrivial properties of the superconducting phase of the kagome superconductors AV_3Sb_5. We built a Ginzburg–Landau theory of the commensurate pair-density-wave state and found that kinks on its ground-state domain walls carry $1/3$ topological charge. Mapping the superconducting fluctuations onto a J_1–J_2 frustrated XY model on an emergent kagome lattice and solving it at finite temperature with tensor networks, we established a $1/3$-vortex-paired phase, i.e. a charge-$6e$ condensed phase, explaining the anomalous $h/6e$ magnetoresistance oscillations observed experimentally.

Xia-Ze Xu, **Tong-Yu Lin**, Guang-Ming Zhang, Efficient optimization of variational tensor-network approach to three-dimensional statistical systems, Phys. Rev. B 112, 134403 (2025) **Co-first author**

• Variational tensor-network optimization for 3D statistical models requires repeatedly contracting a three-layer network made of a projected entangled-pair operator and projected entangled-pair states, which is computationally expensive. We proposed a split corner-transfer-matrix renormalization group (split CTMRG) contraction scheme for this network, reducing the cost from $O(D^{10})$ to $O(D^9)$ without loss of accuracy. Results for the 3D classical Ising model match the precision of state-of-the-art Monte Carlo simulations, with a significant speed-up over previous variational tensor-network methods.

Xia-Ze Xu, **Tong-Yu Lin**, Guang-Ming Zhang, Equivalence of residual entropy of hexagonal and cubic ices from tensor network methods, Phys. Rev. B 113, 214416 (2026) **Co-first author**

• Whether hexagonal and cubic ices have equal residual entropy is a long-standing open problem in statistical physics: analytically only $S_h \geq S_c$ is proven, while numerically the two are extremely close. We encoded the ice rules explicitly into local tensors, turning the residual entropy into the leading eigenvalue of a transfer operator; we proposed a new perspective based on the normality of the transfer operator and showed with tensor computations that the two entropies are numerically equal.

Tong-Yu Lin, Guang-Ming Zhang, Frustration-induced $1/6$-vortex-paired phase in a kagome XY model, arXiv:2610.41161, submitted to Phys. Rev. Lett. **First author**

• Frustration can fractionalize integer vortices in 2D XY systems, but a given model usually hosts only one type of fractional vortex. Studying a J_1–K XY model on the kagome lattice, we proposed a mechanism in which competing frustrated manifolds give rise to new fractional-vortex-paired phases, realizing $1/2$-, $1/3$- and $1/6$-vortex-paired phases within a single model.

Tong-Yu Lin and Guang-Ming Zhang, General tensor network theory for frustrated classical spin models in two dimensions (II): thermal order by disorder, in preparation **First author**

• A follow-up to the general tensor-network theory for frustrated classical spin models: we use the method to study thermal order by disorder in 2D systems, and determine whether a system exhibits thermal order by disorder via an approximation based on local tensors.

Honors and Awards

First-Class Scholarship, Department of Physics, Tsinghua University (2025, 2026) Outstanding Graduate, Nanjing University (2021)
First-Class Scholarship, Department of Physics, Nanjing University (2019)